



IEEE International Conference on
Web Services

Context-aware Holistic Service Policy Enforcement across Swarm-Edge Continuum

Hong-Tri Nguyen, Liang Yuan, Anh-Dung Nguyen, M. Ali Babar, Hong-Linh Truong
Aalto University and Adelaide University

hong-tri.nguyen@aalto.fi

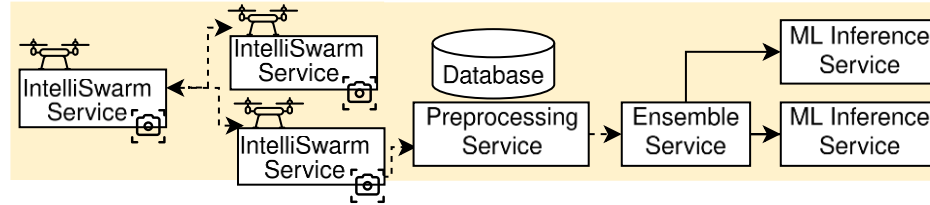
July 11, 2025

Outline

1. Scenarios and motivation
2. Policy adaptation framework for swarm-edge continuum
3. Experiments

Motivation: Service-based Swarm-Edge applications

- Swarm-edge infrastructures support emerging applications requiring mobility, responsiveness



- From **tightly coupled** software-hardware to **decoupled** service-infrastructure for the swarm-edge continuum
 - Proposal: Swarm-Edge-cloud Service-based Application (SESA)
 - Goal: Decouple infrastructure and logic for flexible, adaptive orchestration

Challenges and open problem:

How to support dynamic policy enforcement from the context-based system changes in swarm-edge infrastructure?

Concerns to build a policy adaptation framework

- C1: Which **policy languages** are best suited for swarm-edge setting, when considered from a policy continuum perspective?
- C2: Which **policy coordination** mechanisms are most suitable for managing services of across swarm-edge infrastructures in various situations?
- C3: How do different **situations** impact policy enforcement?

Concern decision in a policy adaptation framework for SESA

C1: Policy language

- Rego with OPA
- Zanzibar/OpenFGA
- Cedar



Rego with OPA

C2: Policy coordination

- Centralization
- Hierarchical centralization
- Hierarchical decentralization



Centralization

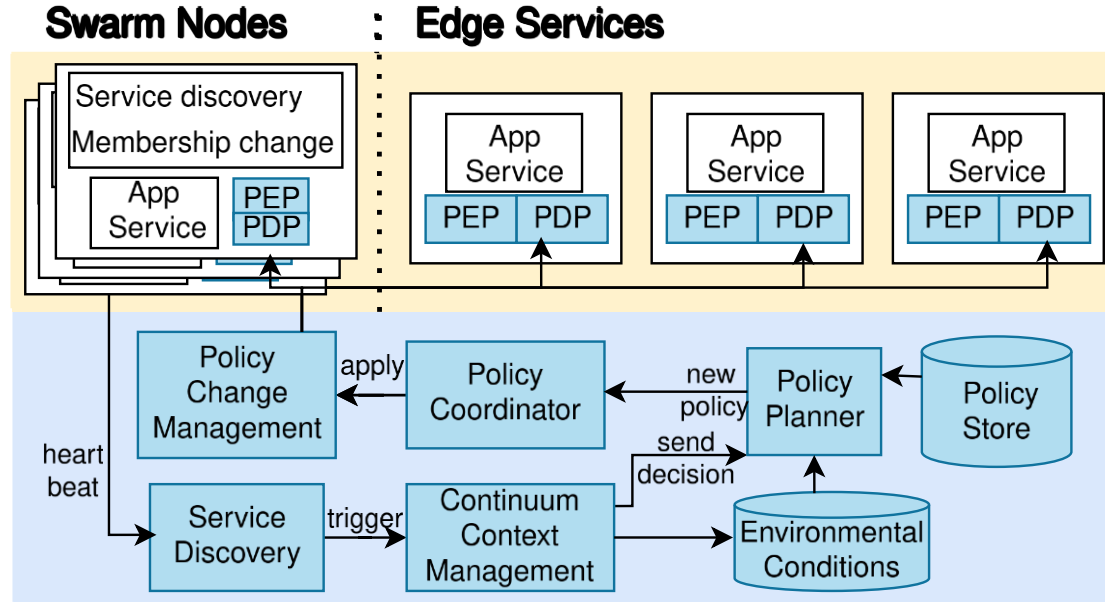
C3: Situations for policy enforcement

- Swarm-edge infrastructures
- Adjustments of application services
- Elasticity changes

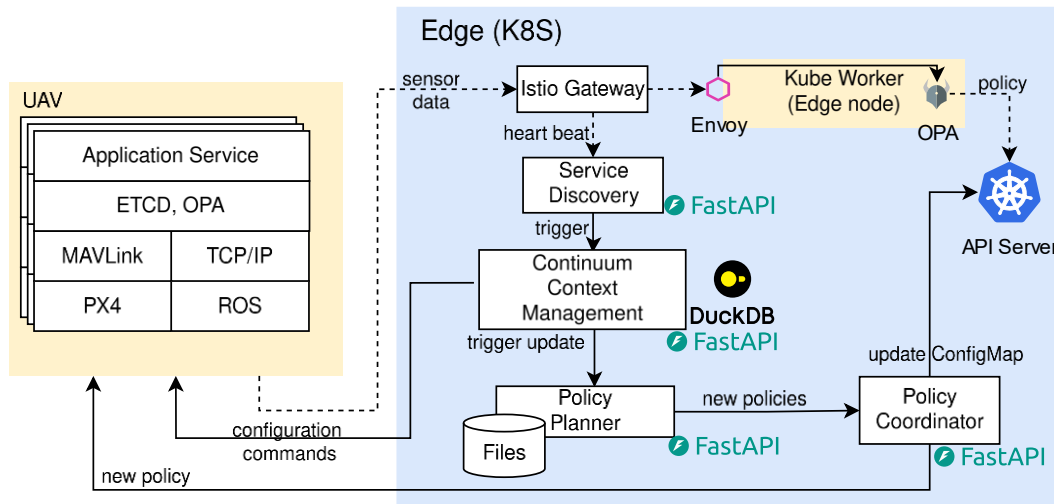


Swarm-edge infrastructures

SwArm-edGE PoLicY (SAGELY)



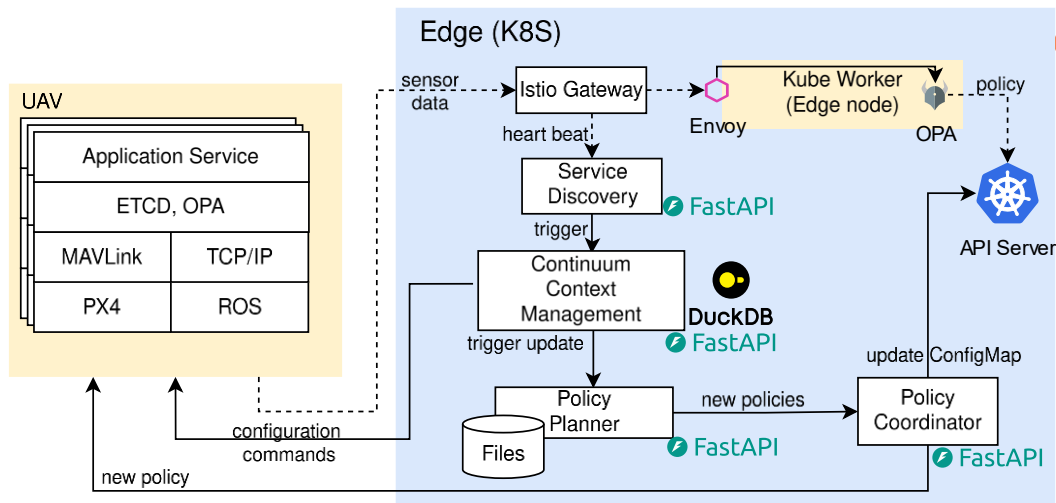
A SAGELY prototype



Setting infrastructures

- UAV stack: PX4, ROS
- Edge stack: Kubernetes and Istio
- Policy:
 - PDP: OPA with Rego
 - PEP: Envoy
- Application: object-classification
 - 4 services
 - 2 inference models from ONNX

A SAGELY prototype



Policy specification and enforcement

- A new Rego-based policy is structured via the policy planner
- Edge services use kubelet agent to watch policy changes (configMap) and propagate to OPA volumes
- Policies are serialized and sent from edge to UAV over MAVLink (UDP)

Experiments

Experiment goal: understanding the impact of policy updates

- Latency of policy propagation in an edge cluster
- Latency of policy propagation to a single UAV and 3-UAV

Experiment settings

- Edge cluster: 4 e2-standard-8 (each machine spec is 8 vCPU, 32 GB RAM, 16 Gbps)
- UAV: HPZ240 (8 CPU, 16 GB RAM, 1 Gbps)
- 3-UAV: 3 Beelink BT3 (each machine x86, Atom x5-Z8300 4 CPU, 4GB RAM 1 Gbps)

Purpose	Machine	Arch	Cores	RAM
Edge	e2-standard-8	x86	8CPU	32GB
UAV	HP Z240	x86	8CPU	16GB
UAV	Beelink BT3	x86	4CPU Atom x5-Z8300	4GB

Latency of policy propagation in an edge cluster

- Service sizes: 4-service, 12-service, 40-service applications
- Policy sizes: 3Kb, 30Kb, 300Kb

Policy size	Latency (ms)		
	4 services	12 services	40 services
3 Kb	567.49	750.54	954.29
30 Kb	750.89	745.41	1421.68
300 Kb	2694.86	3650.91	10090.99

Discussion

Policy distribution becomes a bottleneck at scale

- Latency increases with policy size across all service configurations
- Increasing the number of services generally increases latency (not linear)
- Adding more services does not always cause a significant latency rise.

Latency of policy propagation to a UAV and 3-UAV

Settings

The edge (a Jetson AGX Xavier) updates policies using UDP-based MAVLink to

- A UAV (HPZ240)
- A three-UAV swarm (three-Beelink)

Parameters

- Policy sizes (3Kb, 10Kb, 30Kb)
- Network conditions
 - normal: no additional network impairments
 - low packet loss: 100 ms $\pm$ 20 ms delay with 1% packet loss,
 - medium packet loss: 100 ms $\pm$ 20 ms delay with 5% packet loss,
 - high packet loss: 100 ms $\pm$ 20 ms delay with 10% packet loss.

Latency of policy propagation to a UAV and 3-UAV

A HPZ240 UAV setting						
Network issues (delay + loss)	3Kb		10Kb		30Kb	
	Latency	Err	Latency	Err	Latency	Err
normal	167.88	0%	467.98	0%	1422.22	0%
100ms±20+1%	267.61	2%	567.61	2%	1517.48	0%
100ms±20+5%	270.22	3%	577.93	4%	1518.75	1%
100ms±20+10%	267.57	10%	574.09	7%	1521.62	11%
Three-Beelink setting						
normal	305.86	0%	815.76	0%	2503.67	0%
100ms±20+1%	407.42	1%	903.25	0%	2399.42	1%
100ms±20+5%	414.36	4%	910.33	4%	2483.77	5%
100ms±20+10%	400.55	12.3%	873.81	11%	2375.96	7.3%

Conclusion

Swarm-Edge-cloud Service-based Application (SESA)

- Decoupled service-infrastructure for the swarm-edge continuum

SwArm-edGE PoLicY (SAGELY)

- A context-aware policy adaptation to adjust policy enforcement
- A pluggable service architecture for continuum context management and policy planning

Future work

- Improving adaptive retransmission and error resilience to mitigate propagation failures
- Evaluating hierarchical policy distribution
- Towards a full swarm-edge-cloud continuum architecture

Thank you
Q&A